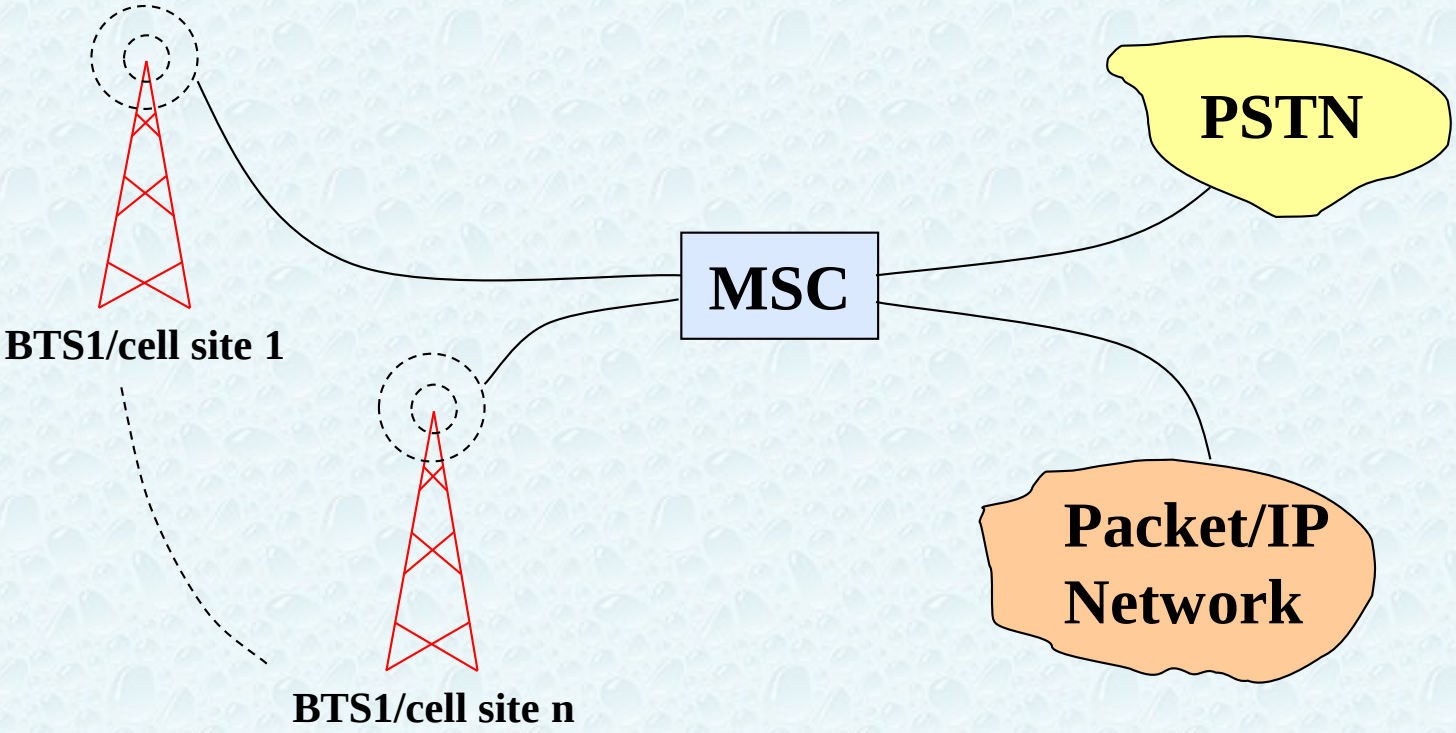
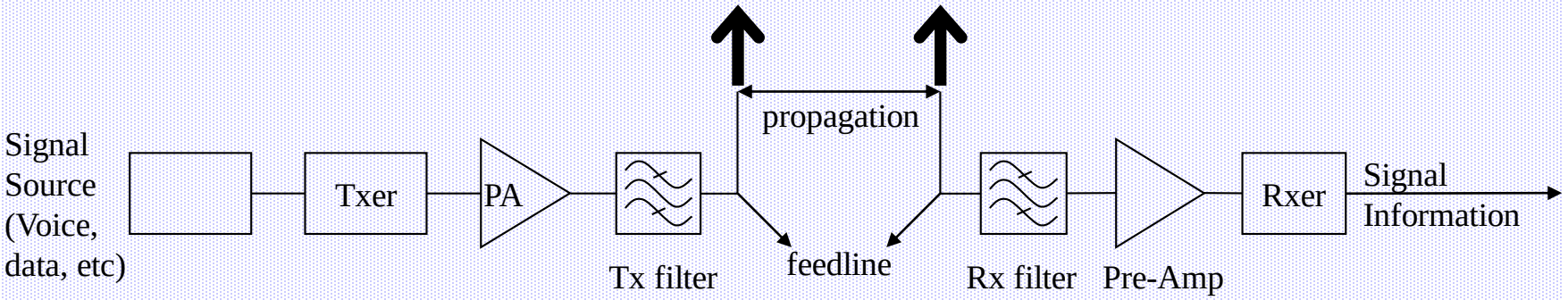
Radio Aspects, Cell Sites and Antenna Subsystem

Cellular Radio Access System

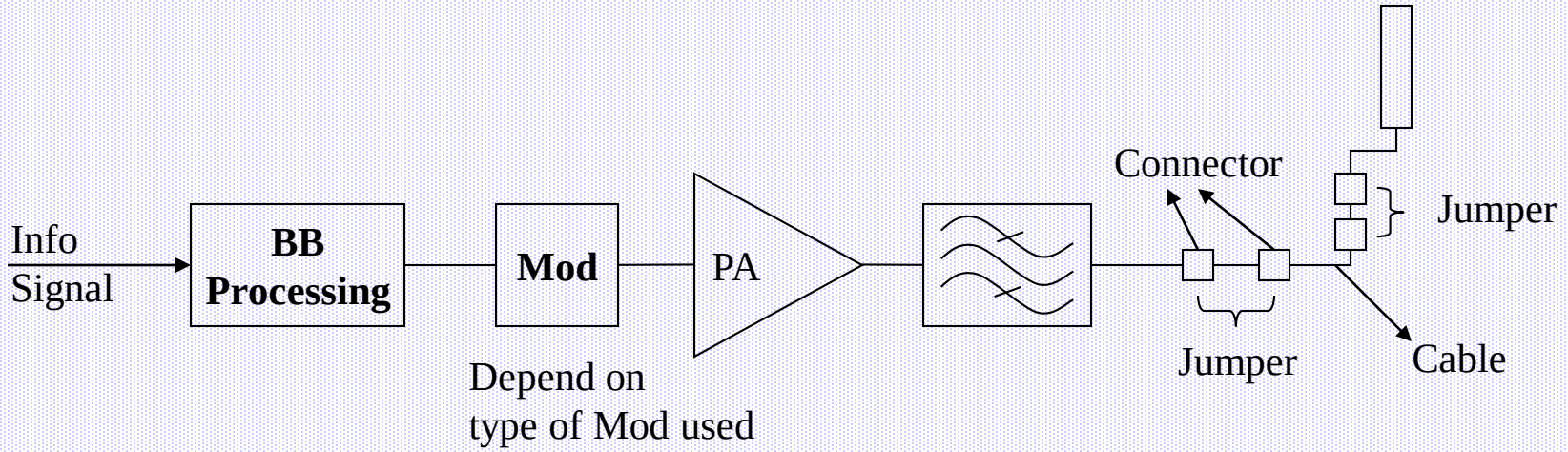


Radio (Tx & Rx) System



- Signal Source: Informasi & Baseband Processing.
- Tx-er: Modulator, Channel Encoder, Interleaver, etc.
- PA: Power Amplifier.
- Feedline: Cable, Connector and Jumper.
- Pre-Amp: LNA.
- Rx-er: Demodulator, Channel Decoder, De-Interleaver, etc.

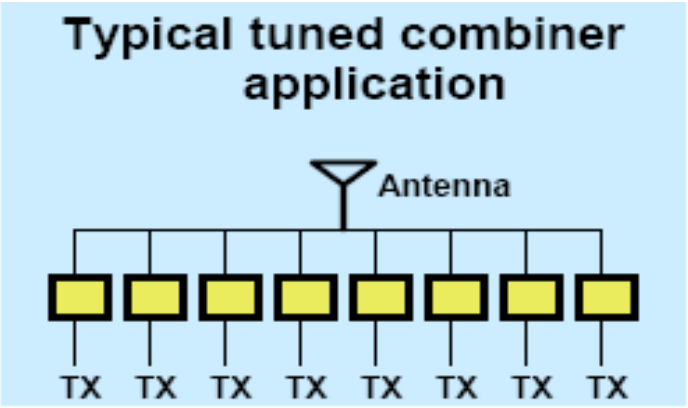
Structure of Transmitter

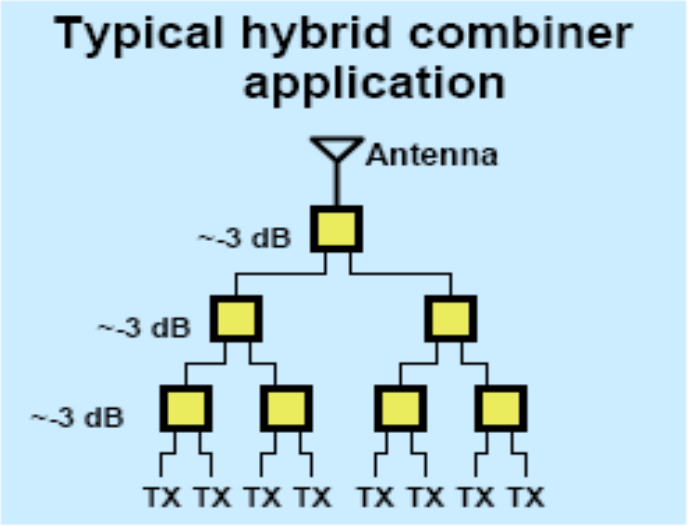


- BB Processing: to process analog signal into digital signal & other processing
- Mod: translate from BB freq. To RF freq → depend on type of cellular system being used e.g. G-MSK modulator for GSM.
- Power Amp:
  - Class A: high linearity
  - Class B: greater output power more efficient than Class A, but less linear
  - Class AB: combined adv. of class A & B → become widely used in wireless.
  - Class C: more power efficient → widely used in wireless

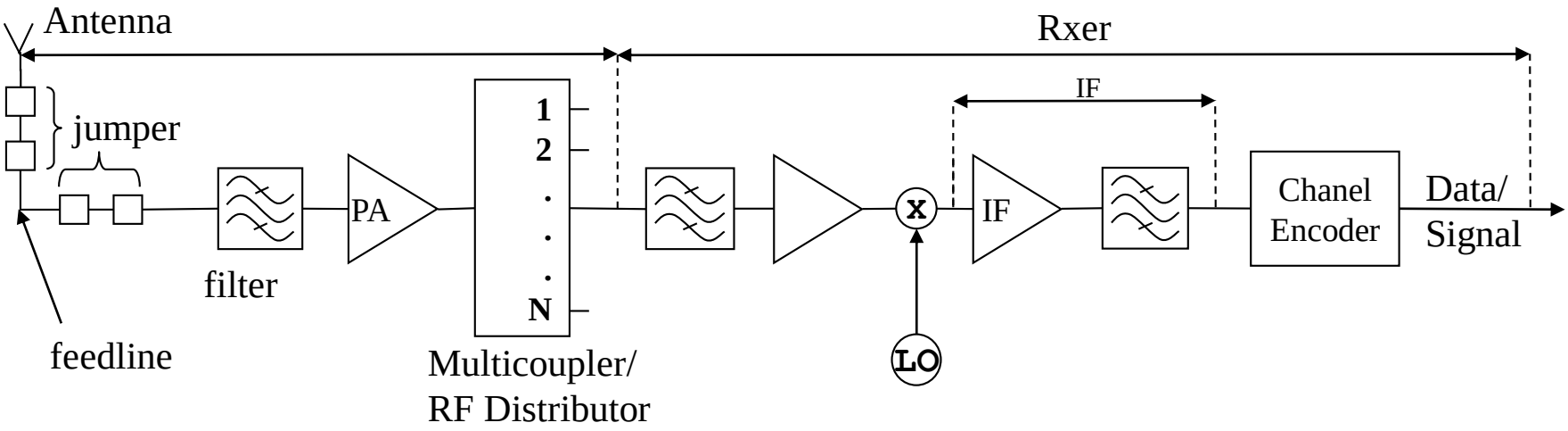
Transmitting Combiners

- Allows multiple transmitters to feed single antenna, providing
  - Minimum power loss from transmitter to antenna
  - Maximum isolation between transmitters
- Combiner types
  - Tuned
    - low insertion loss **~1-3 dB**
    - transmitter frequencies must be significantly separated
  - Hybrid
    - insertion loss **-3 dB** per stage
    - no restriction on transmitter
- frequencies
  - Linear amplifier
  - Linearity and intermodulation are major design and operation issues





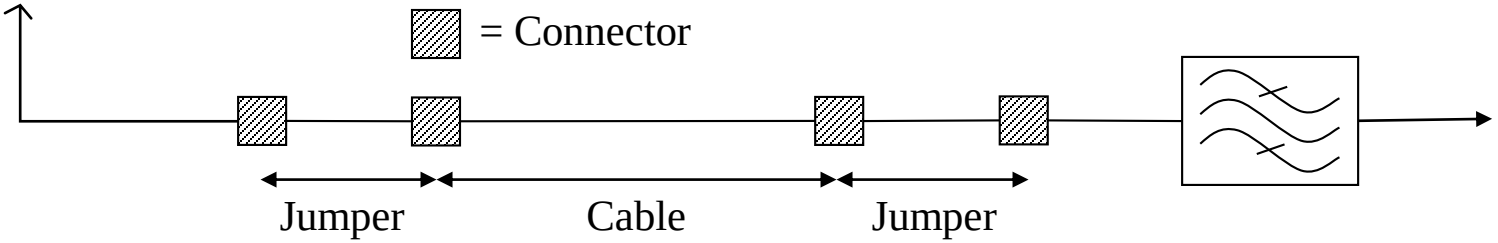
Generic Structure of Rxer



- Block diagram of Rxer varies depend on type of modulation, encoder, and/ or base band processing.
- Parameters to be considered are:
  - frequency range
  - dynamic range
  - sensitivity
  - distortion
  - noise
  - tuning speed

Receiver Components

- Antenna: to convert electromagnetic energy from atmosfer electric energy and transfer it to feed line
- Feed line

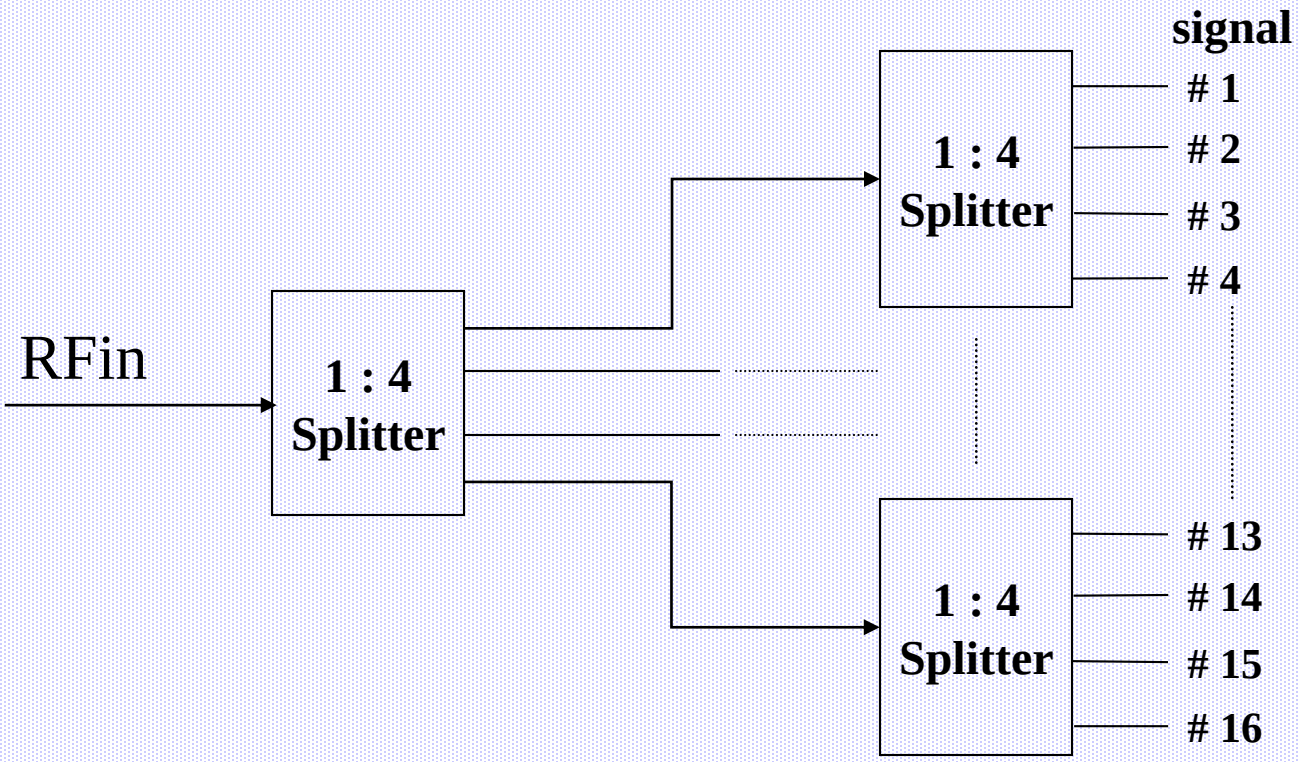


→ *Jumper to ease maintenance and installation*

- Filter & Pre-Amplifier:
  - Filter: to pass the wanted signal & attenuated the interference → designed to work according to the intended bands
  - Pre-Amplifier is used to increased S/N of received signals.

Receiver Components

- Multicoupler:
  - used for RF distribution
  - many signals/users can share the same receive antenna:



Performance Criteria of Receivers

- **Sensitivity:**

  - ability to detect a weak signals, measured by minimum discernible signal (MDS).
  - MDS is measured by turning off the AGC, input a signal with correct BW, and increasing the signal output from generator until S + N = 3 dB higher than 0 when there is no signal.
  - Sensitivity incorporate thermal noise, NF and BW, defined as:

$$\mathbf{S_{en} = 10 \log (kTB) + 10 \log (Channel\ BW) + NF}$$

where: 10 log (kTB) = -174 dBm/Hz for T = 25°C,

B = 840 MHz and k = 1.38 x 10^{-23} J/K

$$\mathbf{S_{en} = -174 + 10 \log(W) + NF}$$

where: W = Channel Bandwidth

e.g. for IS-9 → W = 1.23 MHz

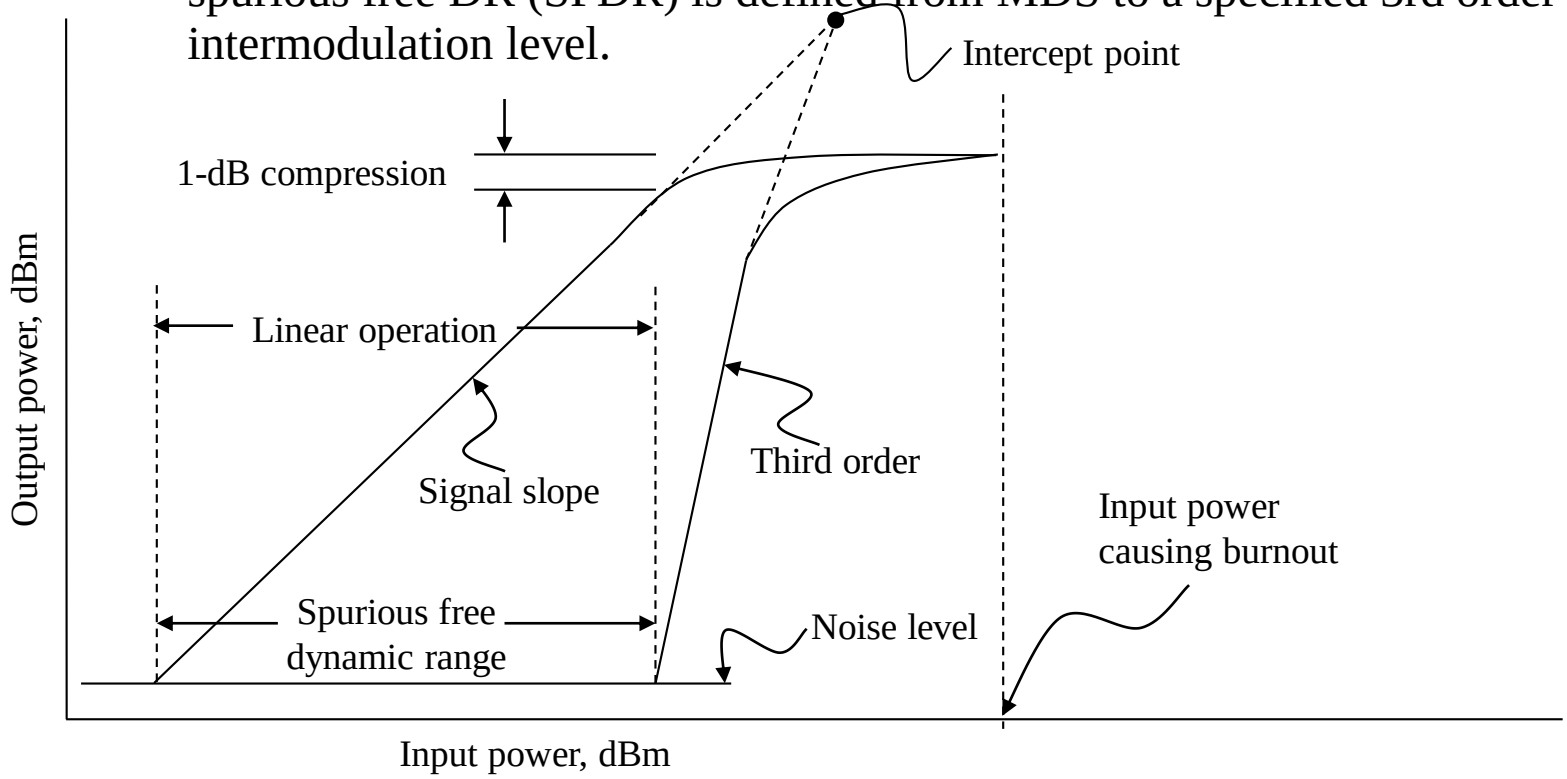
S = -174 + 10 log (1.23 x 10^{6}) + 4 = 109.1 dBm

GSM → W = 200 kHz

S = -174 + 10 log (2 x 10^{5}) + 4 = -117 dBm

Performance Criteria of Receiver

- Dynamic Range
  - a range of levels of the signal that receiver can handle accurately.
  - blocking DR is defined from MDS to 1 dB compression point.
  - spurious free DR (SFDR) is defined from MDS to a specified 3rd order intermodulation level.



- e.g. a range from -13 to -104 dBm → DR = 91 dB

Performance Criteria of Receiver

- SINAD = signal to noise and distortion:

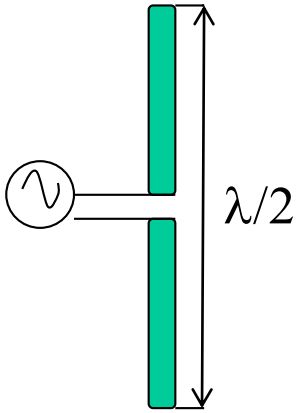
$$SINAD = \frac{S + N + D}{N + D} \quad (dB)$$

- Noise = thermal noise + other noises:
  - → affect overall performance of receiver
  - → quantified by Noise Figure, NF:

$$NF = 10 \log \frac{\left(S/N\right)_{input}}{\left(S/N\right)_{output}}$$

- Selectivity:
  - a measure of protection from off channel interference.
  - depend upon filtering.
  - greater selectivity means better rejection to unwanted signal however if too selective, the signal could be distorted.

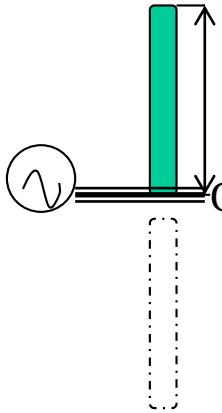
4 Basic Antenna System



G=2.14 dBi

Z= 73 Ω

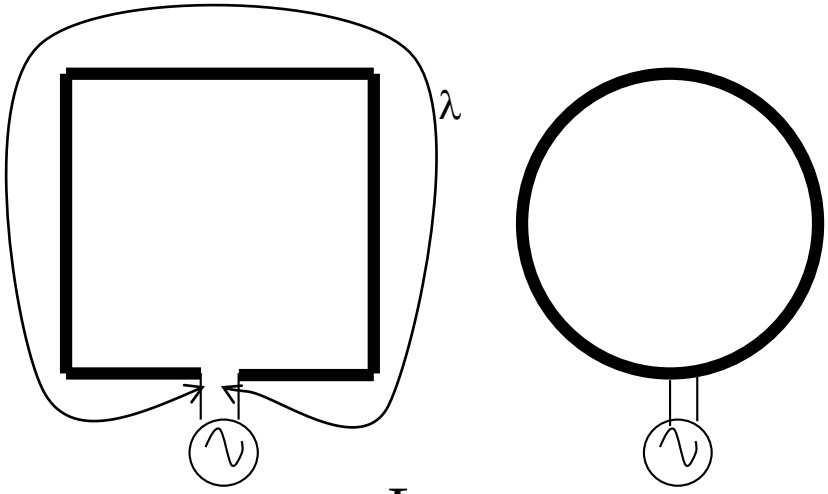
a. Dipole



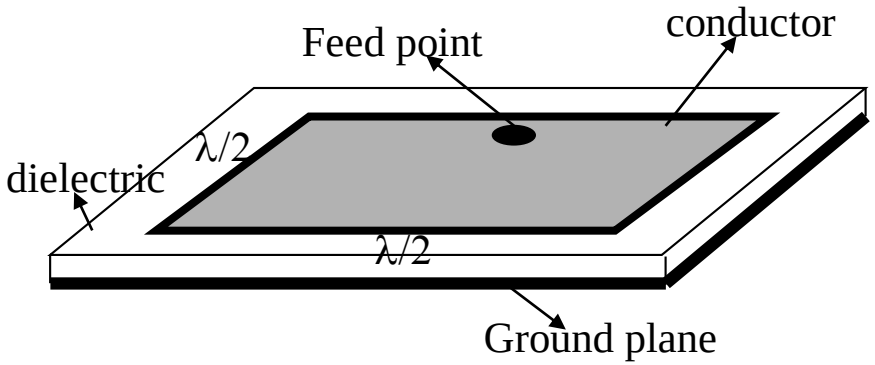
G=4 dBi

Z= 36 Ω

b. monopole



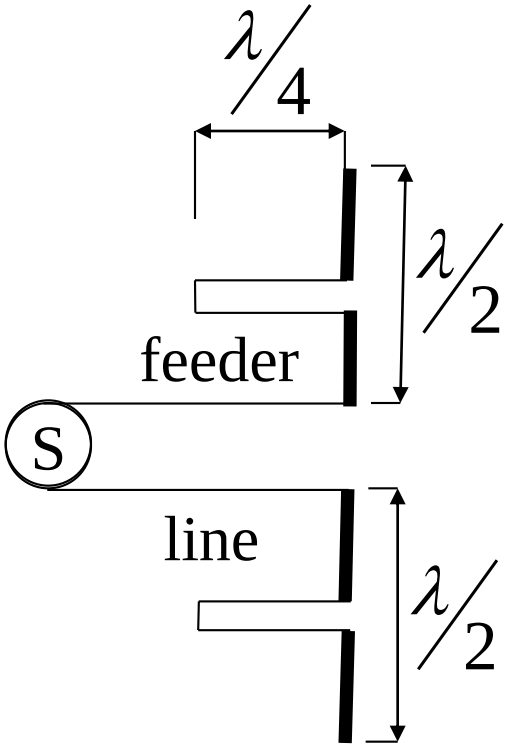
c. Loop

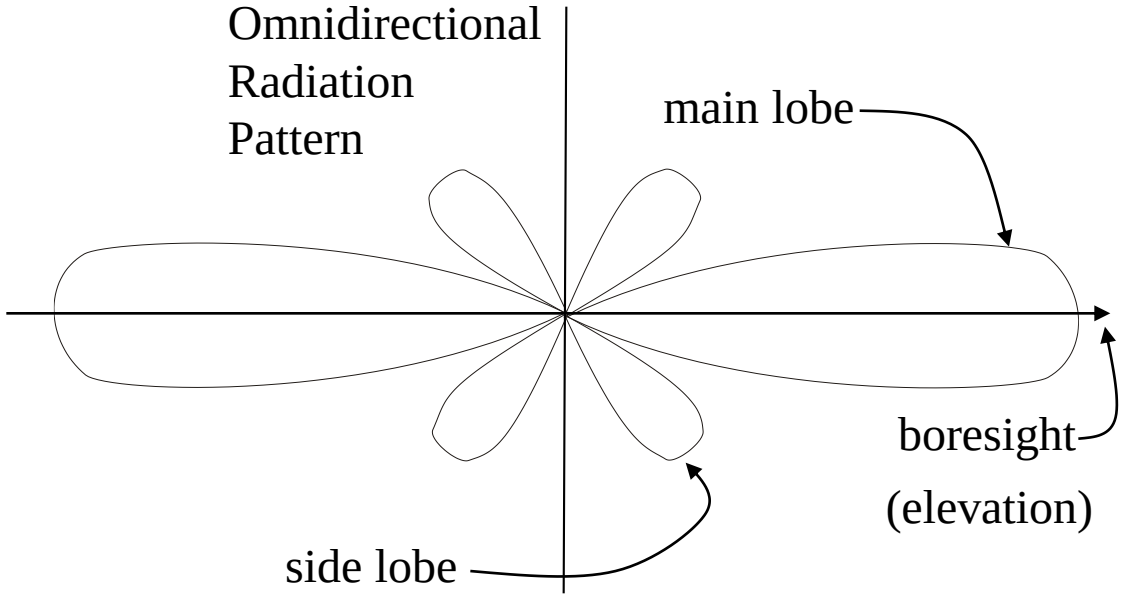


d. Microstrip/ patch

Base Station Antenna

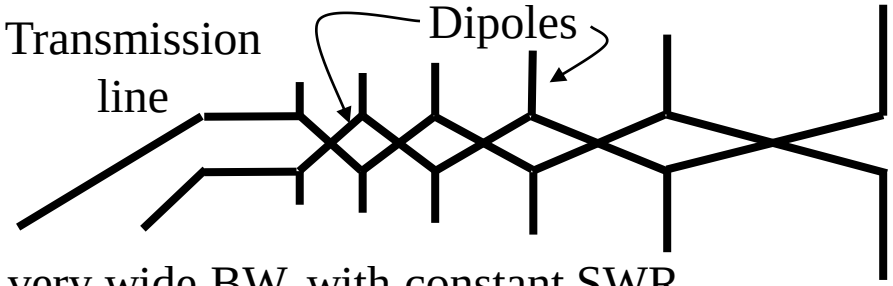
- Use antenna with higher gain
- Could be omnidirectional or sectoral depending on cell type
- Collinear antenna:

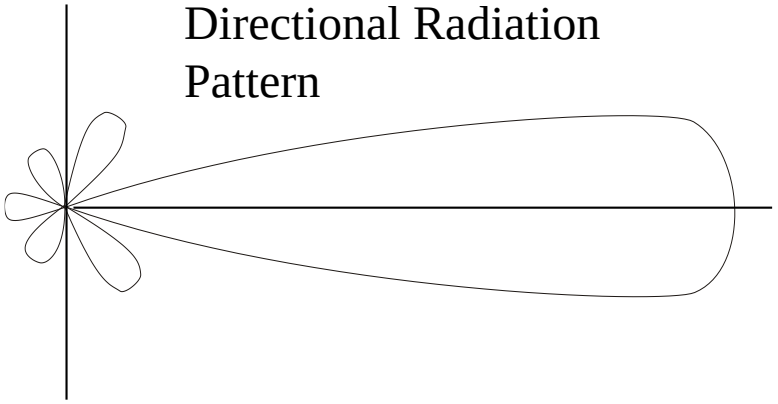




Base Station Antenna

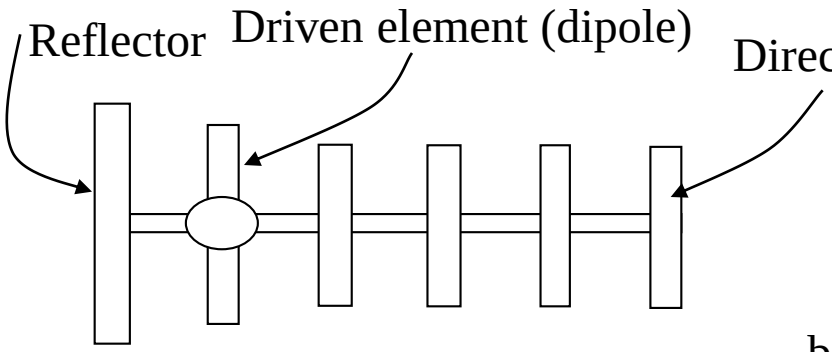
- **Log periodic dipole array (LPDA)**

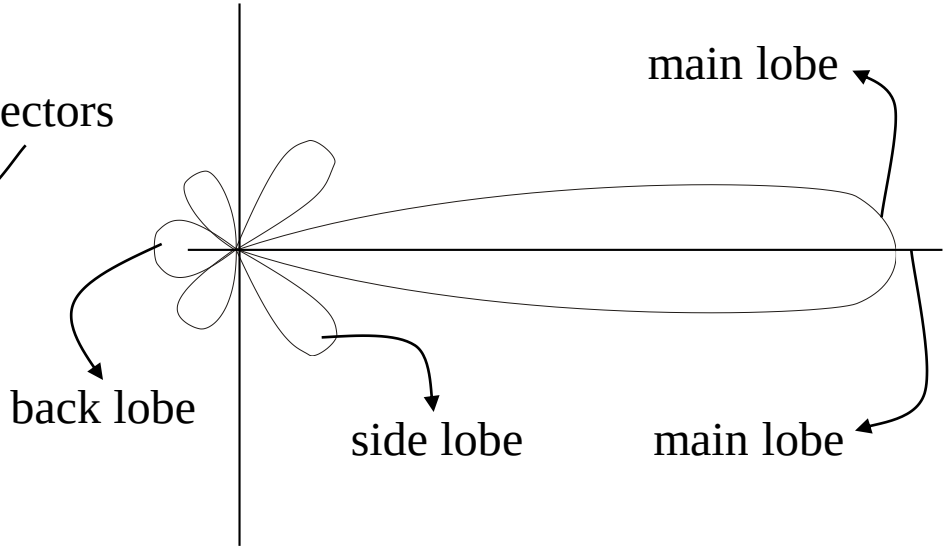




- very wide BW, with constant SWR
- typical gain 10 dBi

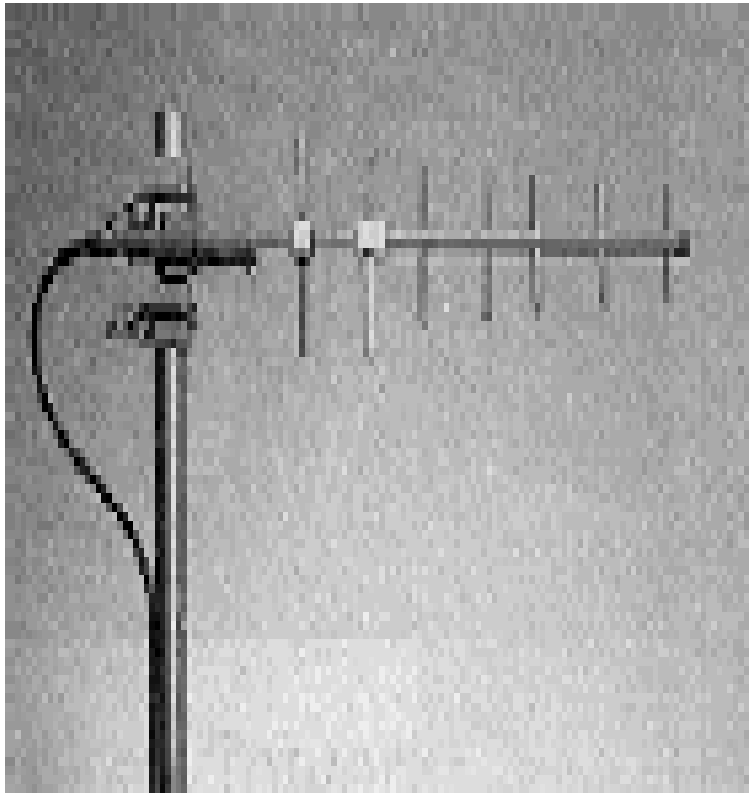
- **Yagi antenna**

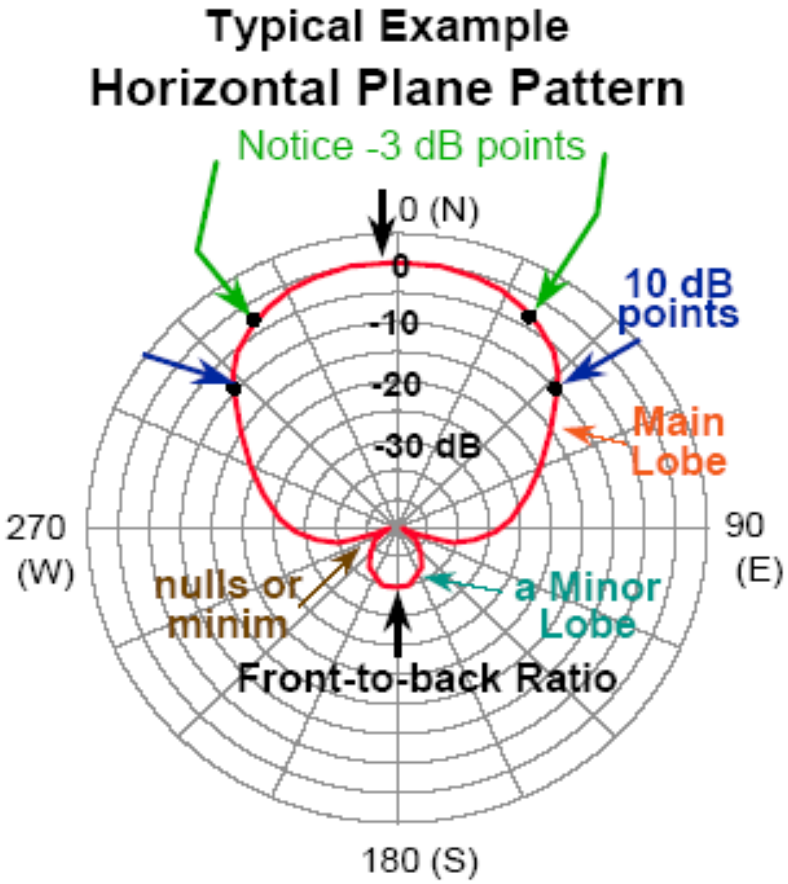




- BW is smaller than LPDA
- typical gain 12 – 14 dB

Directional BS Antenna

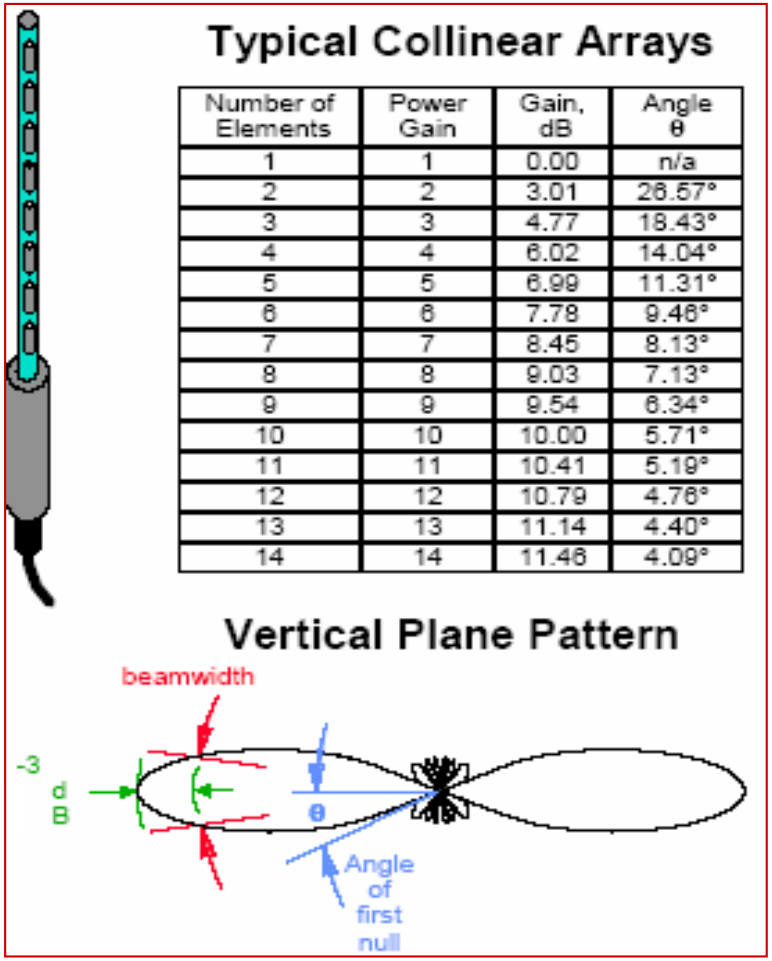




Omni Antennas
Collinear Vertical Arrays

The family of omni-directional wireless antennas:

- Number of elements determines
  - Physical size
  - Gain
  - Beamwidth, first null angle
- Models with many elements have very narrow beamwidths
  - Require stable mounting and careful alignment
  - be sure nulls do not fall in important coverageareas
- Rod and grid reflectors are sometimes added for mild directivity



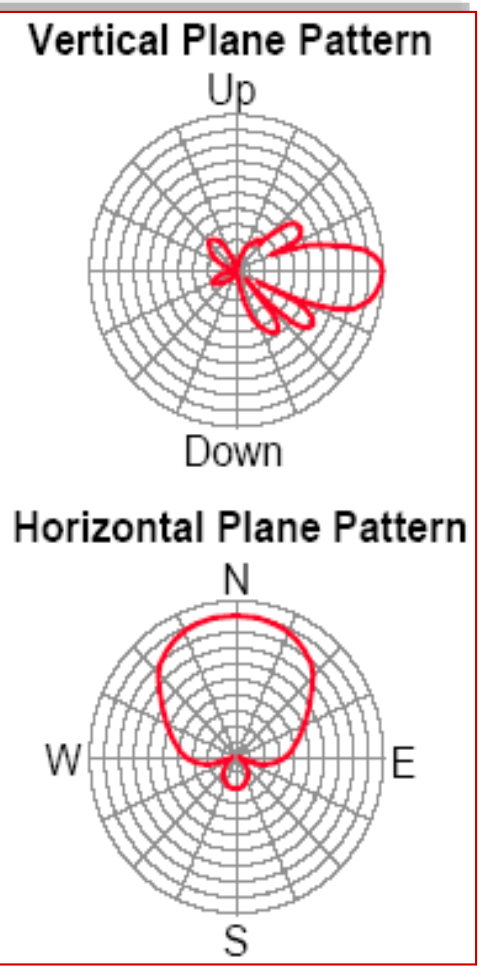
Number of Elements	Power Gain	Gain, dB	Angle θ
1	1	0.00	n/a
2	2	3.01	26.57°
3	3	4.77	18.43°
4	4	6.02	14.04°
5	5	6.99	11.31°
6	6	7.78	9.46°
7	7	8.45	8.13°
8	8	9.03	7.13°
9	9	9.54	6.34°
10	10	10.00	5.71°
11	11	10.41	5.19°
12	12	10.79	4.76°
13	13	11.14	4.40°
14	14	11.46	4.09°

Sector Antennas
Reflectors And Vertical Arrays

Typical commercial sector antennas are vertical combinations of dipoles, yagis, or log-periodic elements with reflector (panel or grid) backing:

- Vertical plane pattern is determined by number of vertically-separated elements
  - varies from 1 to 8, affecting mainly gain and vertical planebeamwidth
- Horizontal plane pattern is determined by:
  - number of horizontally-spaced elements
  - shape of reflectors (is reflector
  - folded?)



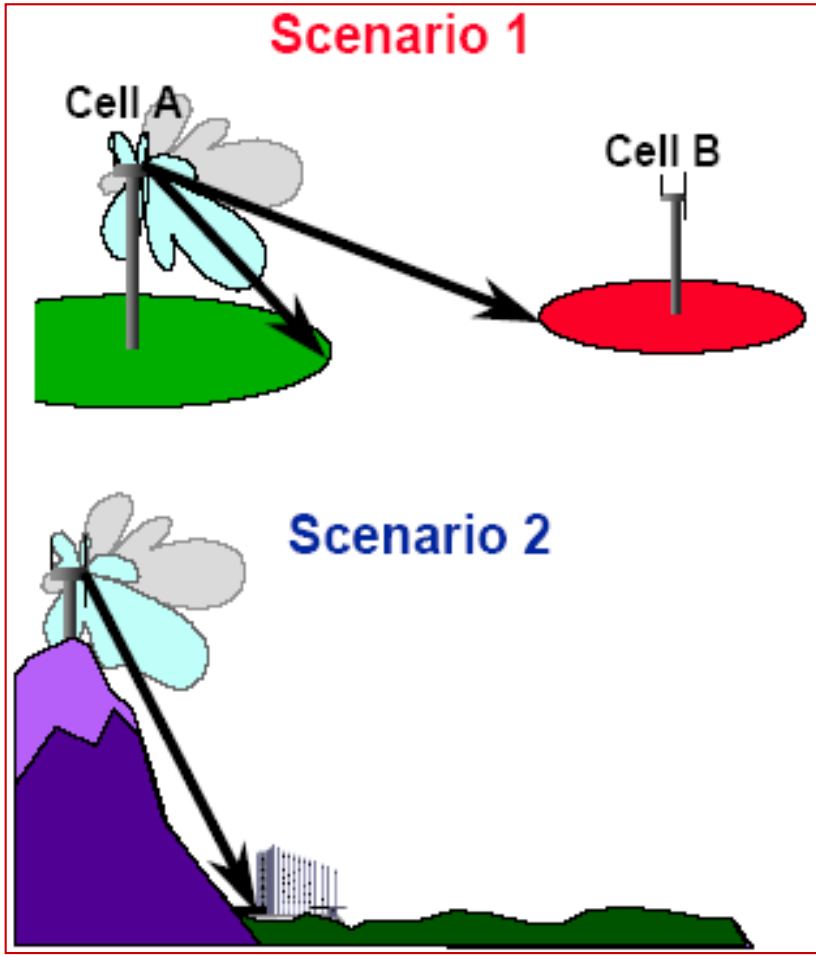


Example Of Antenna Catalog Specifications

Electrical Data			
Antenna Model	ASPP2933	ASPP2936	dB910C-M
Frequency Range, MHz.	1850-1990	1850-1990	1850-1970
Gain - dBd/dBi VSWR	3/5.1 <1.5:1	6/8.1 <1.5:1	10/12.1 <1.5:1
Beamwidth (3 dB from maximum) Polarization	32° Vertical	15° Vertical	5° Vertical
Maximum power input - Watts Input Impedance - Ohms Lightning Protection	400 50 Direct Ground	400 50 Direct Ground	400 50 Direct Ground
Termination - Standard Jumper Cable	N-Female Order Sep.	N-Female Order Sep.	N-Female Order Sep.

Mechanical Data			
Antenna Model	ASPP2933	ASPP2936	dB910C-M
Overall length - in (mm) Radome OD - in (mm)	24 (610) 1.1 (25.4)	36 (915) 1.0 (25.4)	77 (1955) 1.5 (38)
Wind area - ft2 (m2) Wind load @ 125 mph/201 kph lb-f (n) Maximum wind speed - mph (kph)	.17 (.0155) 4 (17) 140 (225)	.25 (.0233) 6 (26) 140 (225)	.54 (.05) 14 (61) 125 (201)
Weight - lbs (kg) Shipping Weight - lbs (kg)	4 (1.8) 11 (4.9)	6 (2.7) 13 (5.9)	5.2 (2.4) 9 (4.1)
Clamps (steel)	ASPA320	ASPA320	Integral

Antenna Downtilt

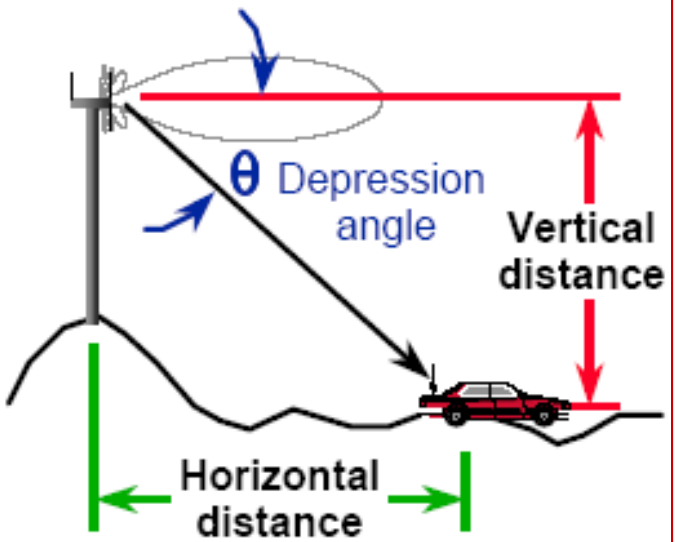


Downtilt is commonly used for two reasons

- 1. Reduce Interference
  - Reduce radiation toward a distant co-channel cell
  - Concentrate radiation within the serving cell
- 2. Prevent "Overshoot"
  - Improve coverage of nearby targets far below the antenna
    - otherwise within "null" of antenna pattern
- ***Are these good strategies?***
- ***How is downtilt applied?***

Vertical Depression Angles

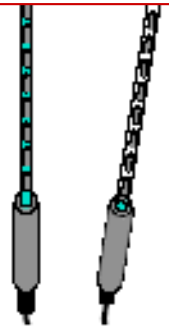
- Basic principle: important to match vertical pattern against intended coverage targets
  - Compare the angles toward objects against the antenna vertical pattern -- what's radiating toward the target?
  - Don't position a null of the antenna toward an important coverage target!
- Sketch and formula
  - Notice the height and horizontal distance must be expressed in the same units before dividing *(both in feet, both in miles, etc.)*



θ = ArcTAN (Vertical distance / Horizontal distance)

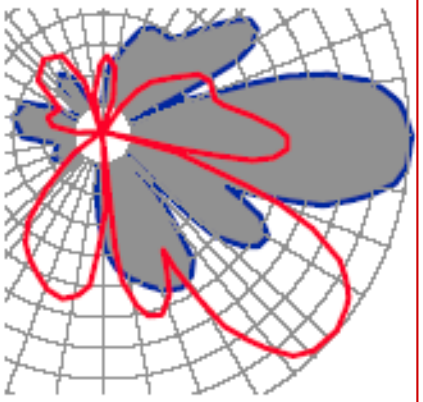
Types Of Downtilt

- Mechanical downtilt
  - Physically tilt the antenna
  - The pattern in front goes down, and behind goes up
  - Popular for sectorization and special omni applications
- Electrical downtilt
  - Incremental phase shift is applied in the feed network
  - The pattern “droops” all around, like an inverted saucer
  - Common technique when downtilting omni cells





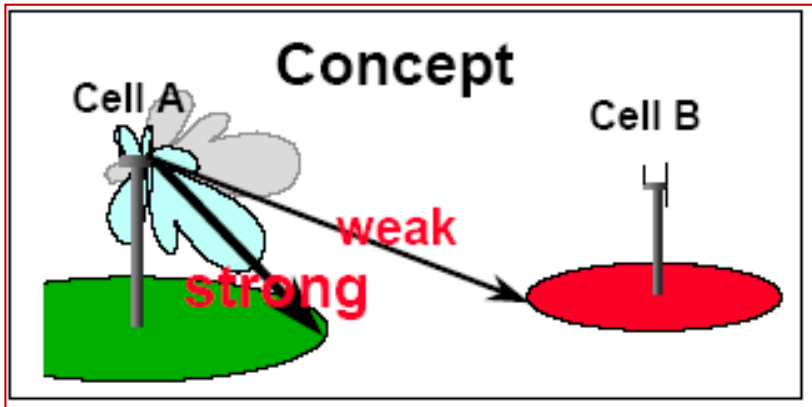


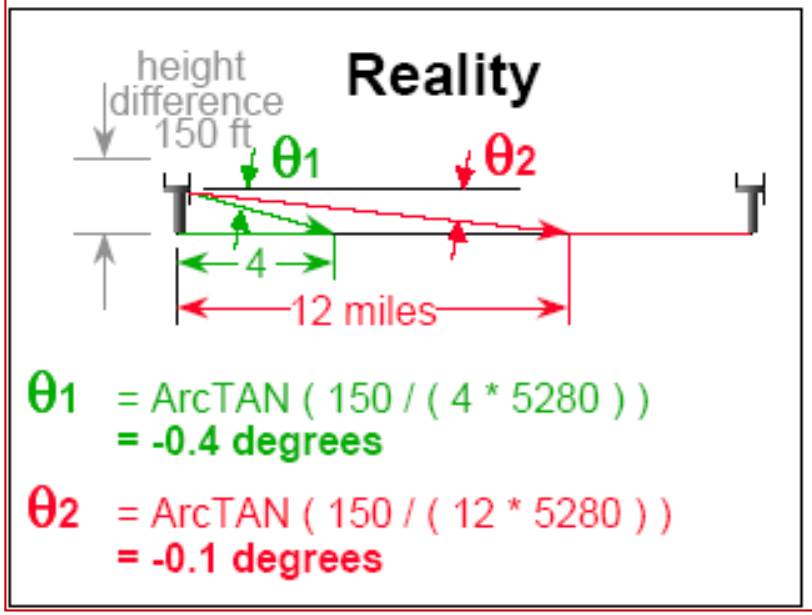






Antenna Downtilt: Reduce Interference





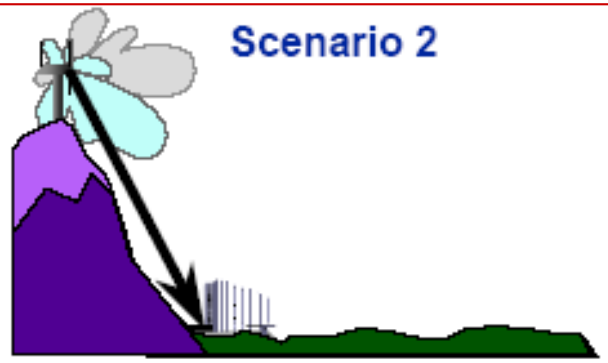
The Concept:

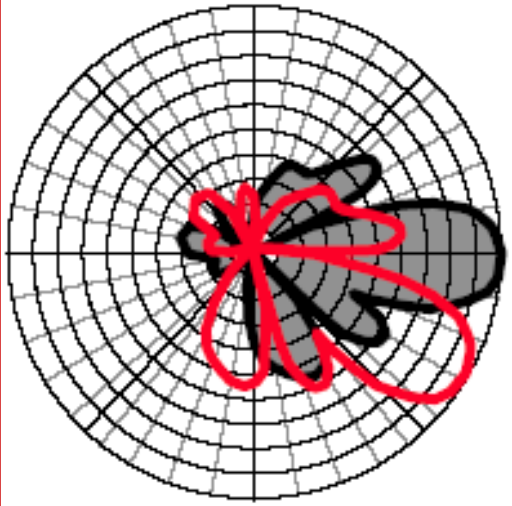
- Radiate a strong signal toward everything within the serving cell, but significantly reduce the radiation toward the area of Cell B

The Reality:

- When actually calculated, it's surprising how *small* the difference in angle is between the far edge of cell A and the near edge of Cell B
  - Delta in the example is only **0.3 degrees**!!
  - Let's look at antenna patterns

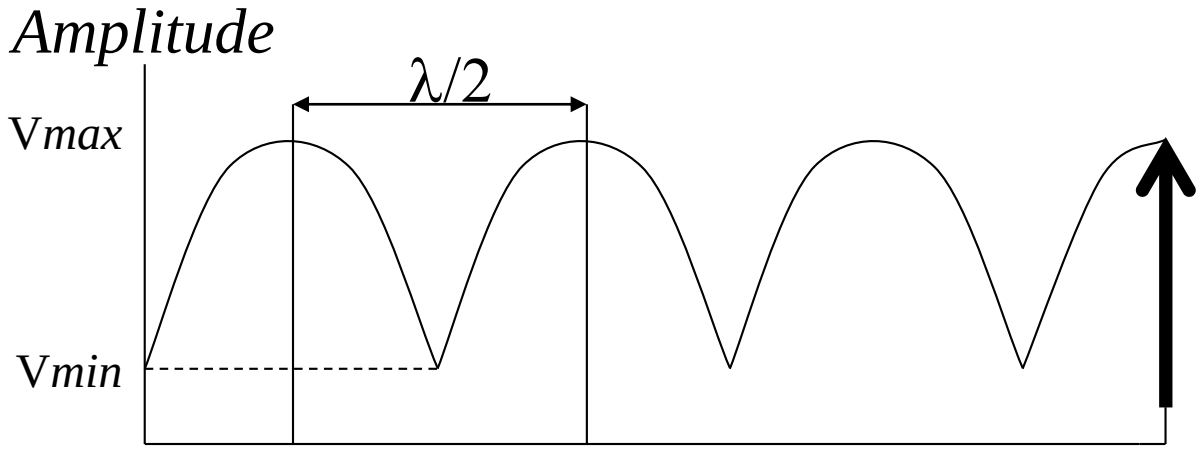
Antenna Downtilt: Avoid Overshoot





- Application concern: too little radiation toward low, close-in coverage targets
- The solution is common-sense matching of the antenna vertical pattern to the angles where radiation is needed
  - Calculate vertical angles to targets!!
  - Watch the pattern *nulls* -- where do they fall on the ground?
  - Choose a low-gain antenna with a *fat* vertical pattern if you have a wide range of vertical angles to "hit"
  - Downtilt if appropriate
  - If needed, investigate special "null-filled" antennas with smooth patterns

SWR of Antenna



- SWR = Vmax/Vmin, define the matching level between antenna and feeder line
- Reflection coefficient: ρ

$$\rho = \frac{SWR - 1}{SWR + 1}$$

where ρ^2 represent a percent of reflected power defined by:

$$\mathrm{Re}\,turn\,Loss = 10\log\left(\rho^2\right)$$

Performance Criteria of Antenna

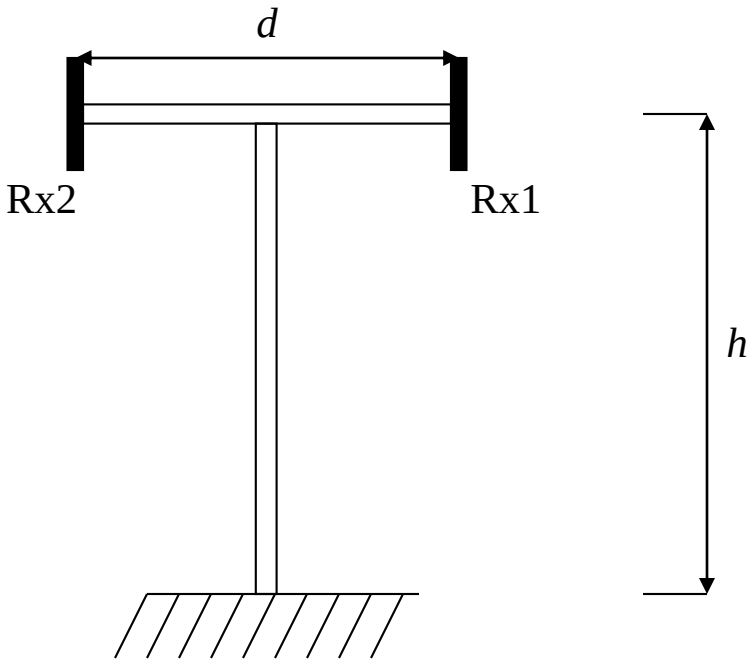
- Antenna pattern, defined at azimuth and elevation orientation → either omni or bidirectional antenna
- Main lobe & side lobe, the lower side lobe the better resistance to interference
- Input impedance, usually complex matching input ipedance and feeder line impedance is very critical to have maximum power transfer from feeder to antenna
- Beamwidth, usually defined as angular separation where there is 3 dB reduction from bore-sight
- Directivity & Gain, is ratio of radiation intensity at wanted direction and coverage radiation intensity over all direction

$$G = \mu.D$$

- Bandwidth, define operating range of antenna, limited by SWR. A typical BW is for SWR 1:1.2 at the band edge.
- Polarization, defined by orientation of $\vec{E}$

Performance Criteria of Antenna

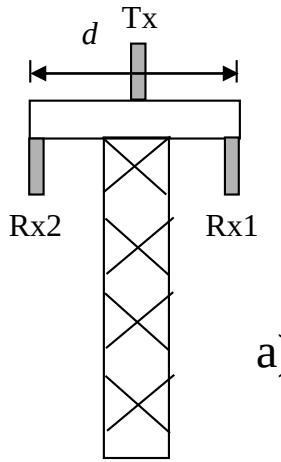
- Front to Back Ratio, is ratio between main lobe & back lobe, very impotant for directional antenna.
- Spatial diversity:



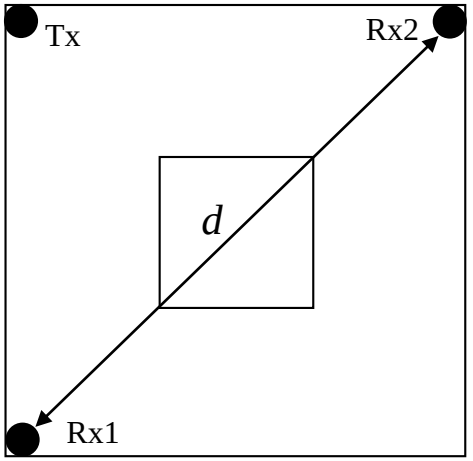
$$d = \frac{h}{11} x \frac{835}{f} \quad (feet)$$

where f is in MHz

Antenna Installation

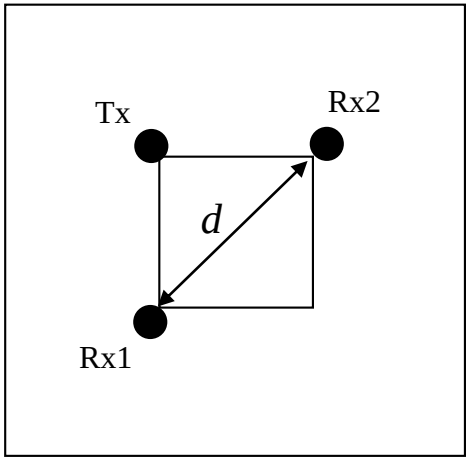


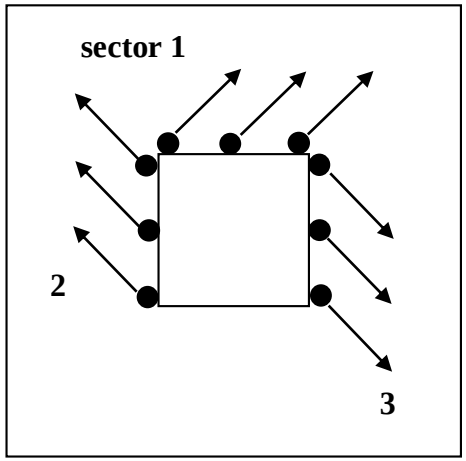
a) Tower

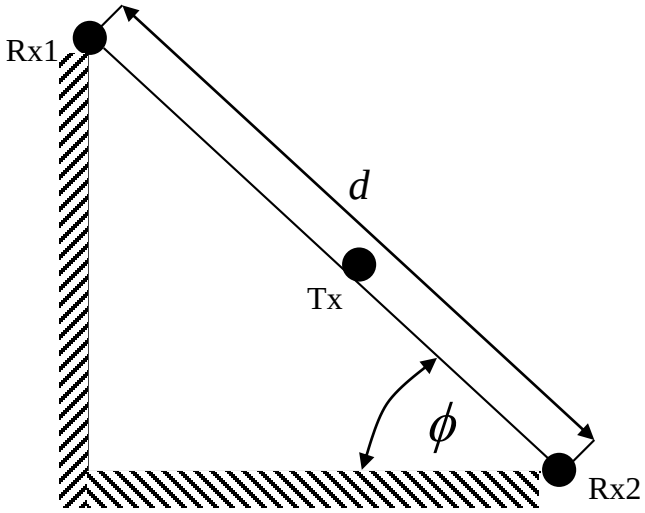


b) Roof Top, Edge of Building

c) Roof Top







d) Wall Mounting

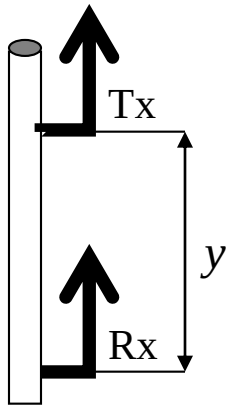
Antenna Installation Tolerance

- Apply to physical oriented & plumbness of its installation
- For omnidirectional antenna, it is unnecessary. But for directional antenna it is very critical
- Usually taken +/- 5% from antenna horizontal/azimuth pattern.

Table: Horizontal Antenna Tolerance

Azimuth/Horizontal Pattern	Tolerance from Bore Sight
110°	+/- 5.5^{o}
92°	+/- 4.5^{o}
60°	+/- 3.0^{o}
40°	+/- 2.0^{o}

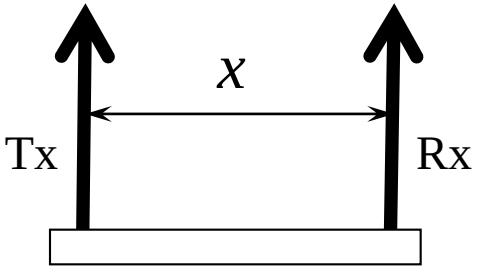
Antenna Isolation



a. vertical

$$VI = 28 + 40\log\left(\frac{y}{\lambda}\right) dB$$

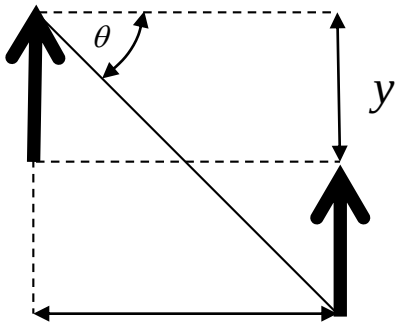
where $y > \lambda$



b) horizontal

$$HI = 22 + 20\log\left(\frac{x}{\lambda}\right) dB$$

where $x > 10\lambda$

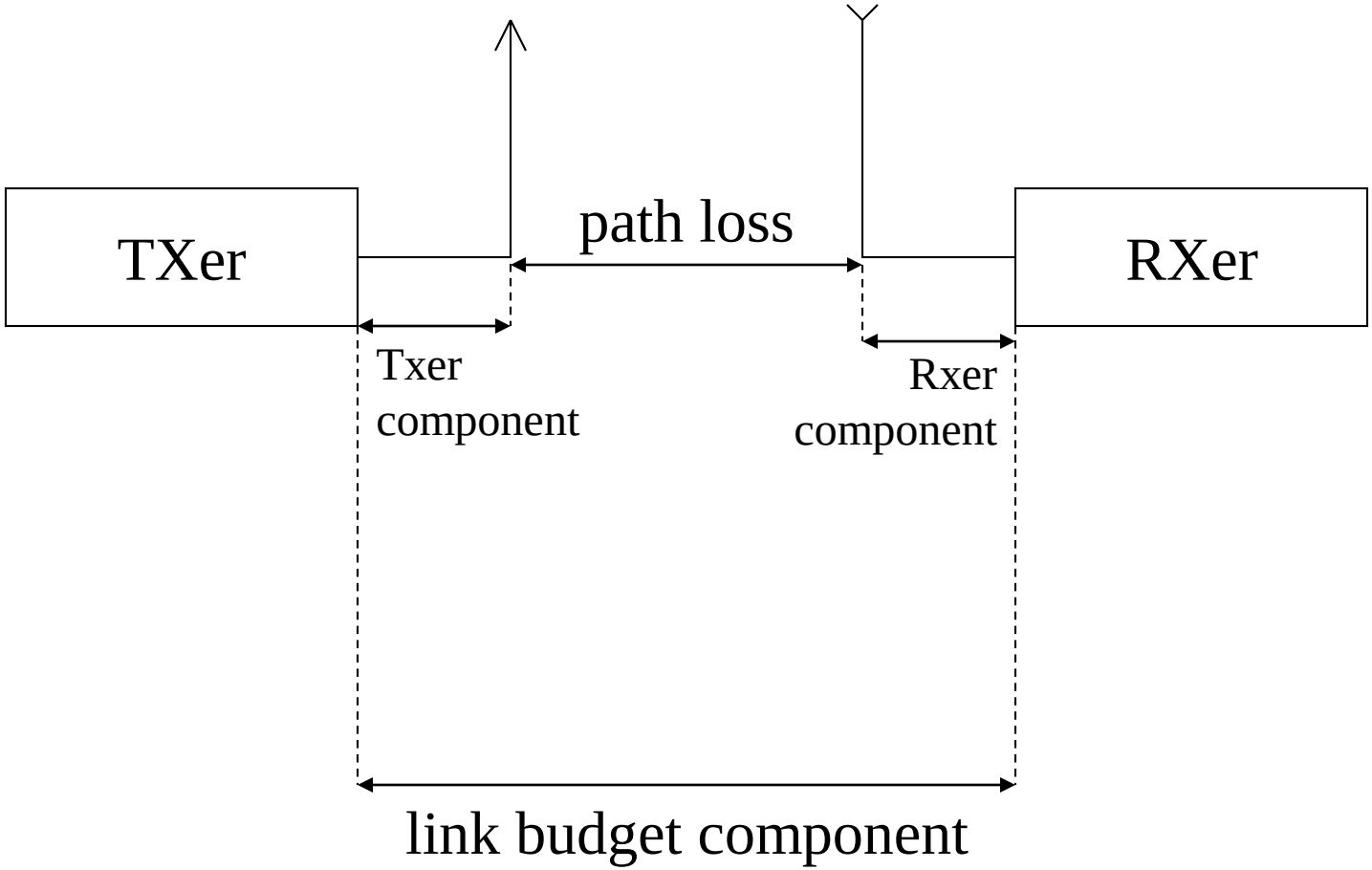


c. slant

$$SI = (VI - HI)\frac{\theta}{90^o} + HI\ dB$$

where θ *slant angle*

Link Budget


TXer
path loss
RXer
Txer
component
Rxer
component
link budget component

Link Budget – Up Link

- **Frequency range, MHz**

- **Mobile parameters**
  - Tx PA output (max)
  - Cable loss
  - Antenna gain

  -------- (Subsc. ERP max, dB)
- **Environmental margins**
  - Fading margin
  - Environmental attenuation
  - Cell overlap

  ---------- (dB)

- **Base station parameters**
  - Rx ant. gain Rx jumper loss
  - Rx tower top amp gain (net)
  - Rx cable loss
  - Rx ligthning arrester loss
  - Rx duplexer loss
  - Rx diversity gain
  - Rx coding gain
  - Rx sensitivity

  ------- Up-link budget, dB

Link Budget – Down Link

- **Frequency range, MHz**
- **Base station parameters**

  - Tx PA output power
  - Tx combiner loss
  - Tx duplexer loss
  - Tx ligthning arrester loss
  - Tx cable loss
  - Tx jumper loss
  - Tx tower top amp gain
  - Tx antenna gain

  (Cell ERP, dB)

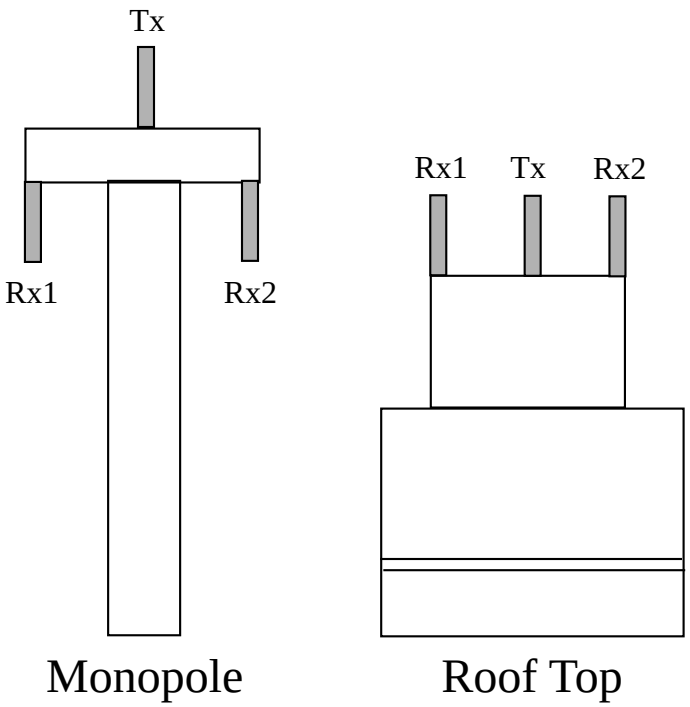
- **Environmental margins**
  - Tx diversity gain
  - Fading margin
  - Environmental attenuation
  - Cell overlap

  (dB)

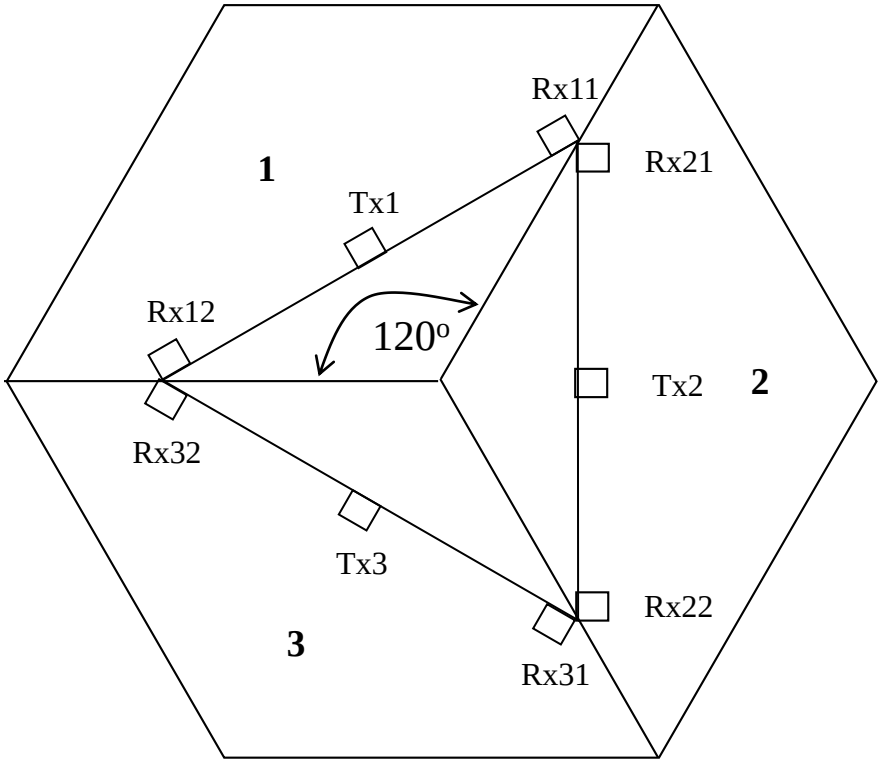
- **Mobile parameters**
  - Antenna gain
  - Rx diversity gain
  - Antenna cable loss
  - Coding gain
  - Rx sensitivity

---------- Down-link budget, dB

Type of Cell Site/BTS (1)

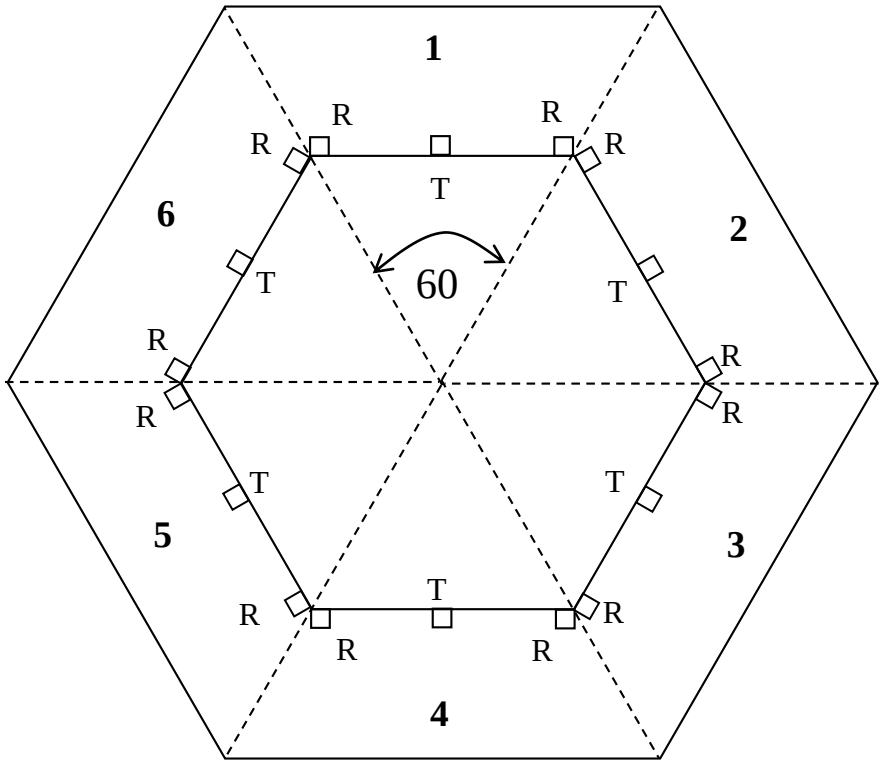


a) Omni cell

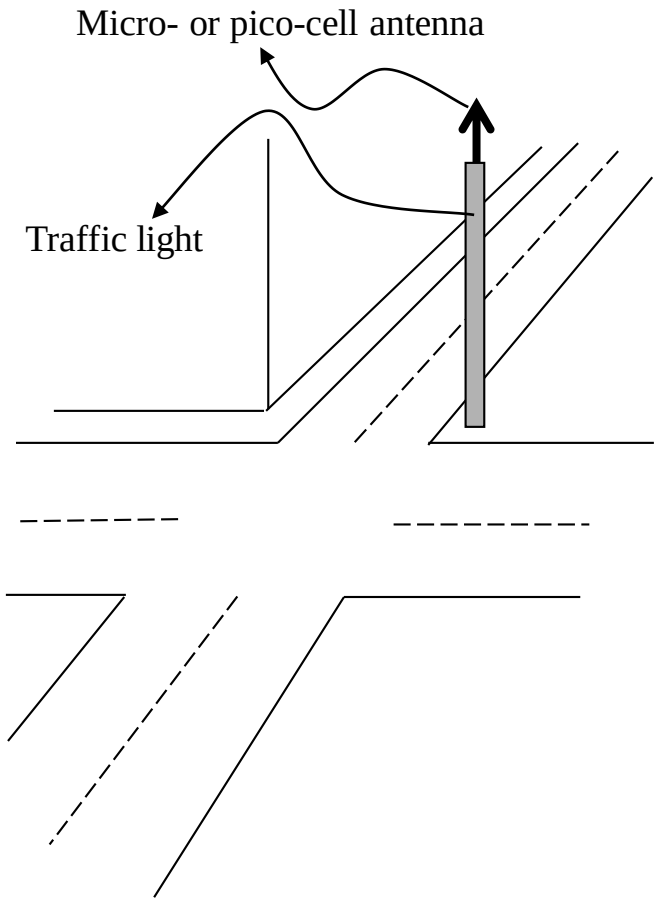


b) 3 Sectors

Type of Cell Site/BTS (2)

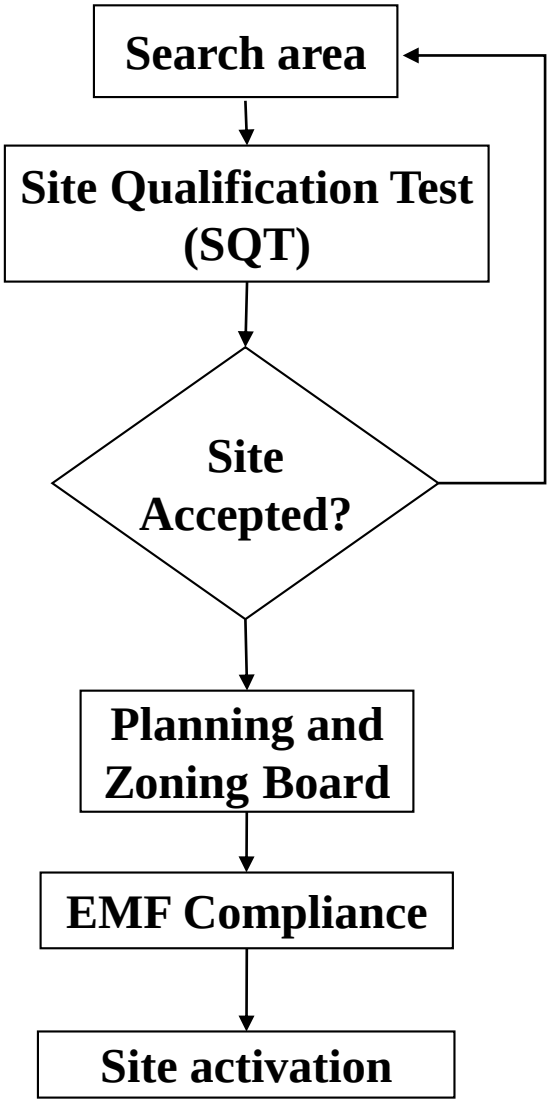


c) 6 sectors



d) Microcell or picocell

Cell Site Design (1)



Cell Site Design (2)

- **Search Area:**
  - searching area to place cell site/BTS that meet the specifications
  - plot the propagation path, including clearance
  - mapping the area for planning & documentation

- **SQT:**
  - to assure the area is a viable candidate for a cell site by measurements
  - include a sketch of the location, antenna type, height, ERP, path clearance, and do callibration

- **Site acceptance:**
  - if SQT is positive then the area is accepted to place a cell site
  - if not, then area is rejected
  - both site acceptance and rejection should be documented

Cell Site Design (3)

- **Planning and zoning board:**
  - why the site is needed
  - how the site will improve the network
  - drawing the sketch of site

- **Electromagnetic Force (EMF) Compliance:**
  - EMF identify the source of EM from the site itself and surrounding area
  - to ensure it complies with personal safety and government regulation
  - incorporated the type of Txer, power, frequency range, etc
  - method for calculating EMF, e.g. IEEE C95.1 – 1991 standard

- **Site activation:**
  - when every steps above is OK, the cell site/BTS could be placed and turn on

Conclusion

- ……
- ……
- ……
- ……
- ……

The End